

MCA SEMESTER-I
School of Computer and Systems Sciences,
Jawaharlal Nehru University,
New Delhi -110067
Computer Fundamentals (CS-402) Mid-I
Date: 07.10.2025

Total Marks: 20

Time: 2hr

All Compulsory

1. a) By means of a timing diagram similar to Fig. 1(a), show the signals of the outputs F1 in Fig.1(b) as functions of the inputs x, y and z.

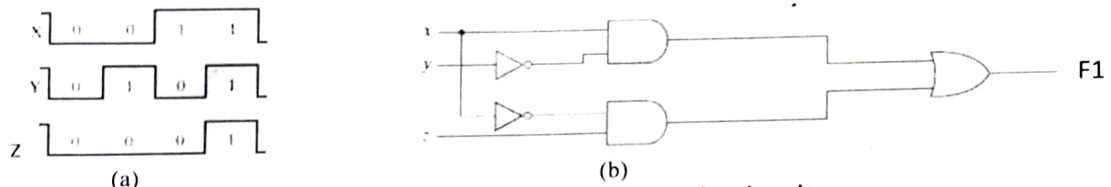
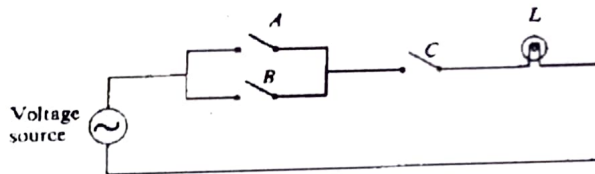


Figure 1: a) Input waveform signals b) Logic circuit

- b) Express the switching circuit in binary logic notation.



$2.5+2.5=5$

2. Design a combinational circuit for BCD-Excess-3 coder converter. 5
3. Design BCD-to-decimal decoder. 5
4. Design 4 to 2 Priority encoder. 5

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Computer Fundamental (CS-402)
End Semester Exam

Total Marks: 50

Date: 08.12.2025

Duration: 3hr

All Compulsory

1. Write note on:
 - a. Basic building block of the logic Gate
 - b. Status Register
 - c. Shift register.
 - d. Accumulator
 - e. Hardwired control and microprogram sequencer control

3x5=15
2. For the following Boolean functions solve accordingly.
 - a. Express the Boolean function $F = xy + x'z$ in a product of sum
 - b. Obtain the truth table of the function F and express each in sum of minterms and product of maxterms. $F = (A' + B)(B' + C)$

2+3=5
3. Simplify the Boolean function F together with don't care condition d , then express the simplified function in sum of minterms.
 $F(A, B, C, D) = \Sigma(1, 3, 5, 7, 9, 15)$
 $d(A, B, C, D) = \Sigma(4, 6, 12, 13)$

5
4. Design and implement full Adder.

5
5. Implement function using multiplexer
 $F(A, B, C, D) = \Sigma(0, 1, 3, 4, 8, 9, 15)$

5
6. Combinational circuit is defined by the function :
 $F_1(A, B, C) = \Sigma(3, 5, 6, 7)$
 $F_2(A, B, C) = \Sigma(0, 2, 4, 7)$
Implement the circuit using PLA.

5
7. a) Explain T-flip flop with logic diagram and characteristic table.
b) Implement 3-bit binary counter using T-Flip flop.

5+5=10